



Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
Institut Teknologi Sepuluh Nopember

Laporan Akhir

Praktikum Jaringan Komputer

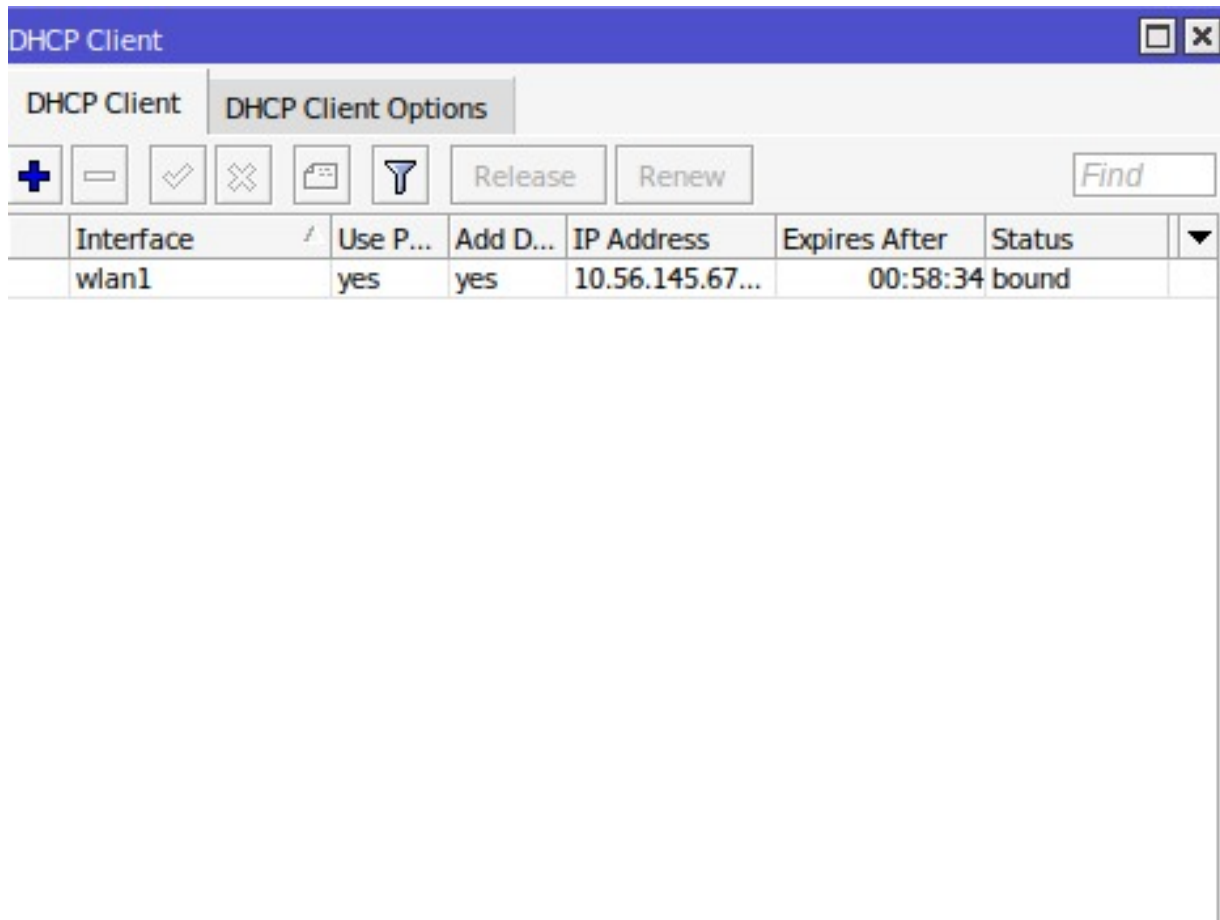
Firewall & NAT

Rayhan Diah Purwadi - 5024241032

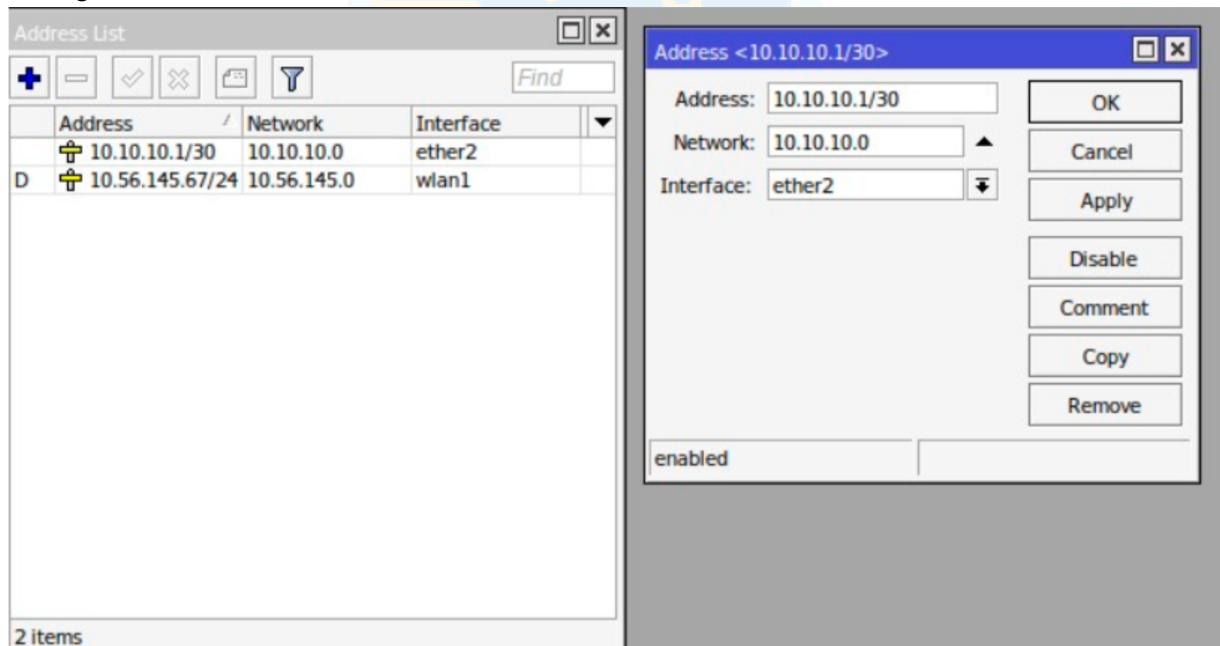
2026

1 Langkah-Langkah Percobaan

1. Lakukan koneksi DHCP ke internet pada hotspot HP.



2. Konfigurasi IP address router A ke router B



3. Konfigurasi routing statis router A

New Route

General Attributes

Dst. Address: 192.168.10.0/24

Gateway: 10.10.10.2

Check Gateway:

Type: unicast

Distance:

Scope: 30

Target Scope: 10

Routing Mark:

Pref. Source:

enabled active

OK Cancel Apply Disable Comment Copy Remove

4. Konfigurasi IP address router B

Address List

Address	Network	Interface
10.10.10.2/30	10.10.10.0	ether1
192.168.10.1/24	192.168.10.0	ether2

5. Konfigurasi routing statis router B

Route List

Dst. Address	Gateway	Distance	Routing Mark	Pref. Source
AS 0.0.0.0/0	10.10.10.1 reachable ether1	1		
DAC 10.10.10.0/30	ether1 reachable	0		10.10.10.2
DAC 192.168.10.0/24	ether2 reachable	0		192.168.10.1

6. Konfigurasi router NAT dengan srcnat di router B

Firewall

#	Action	Chain	Src. Address	Dst. Address	Proto...	Src. Port	Dst. Port	In. Inte...	Out. Inte...	In. Inte...	Out. Inte...	Src. Ad...	Dst. Ad...	Bytes	Packets
0	mas...	srcnat						ether1						0 B	0

7. Konfigurasi IP di PC client router B

```

inet 172.20.0.1/16 brd 172.20.255.255 scope global br-40d5a5d34c27
    valid_lft forever preferred_lft forever
6: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 6a:9d:d7:bd:3e:7f brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
13: enx9c69d378e741: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 9c:69:d3:78:e7:41 brd ff:ff:ff:ff:ff:ff
    inet 192.168.88.2/24 brd 192.168.88.255 scope global noprefixroute enx9c69d378e741
        valid_lft forever preferred_lft forever
    inet 192.168.10.2/24 brd 192.168.10.255 scope global noprefixroute enx9c69d378e741
        valid_lft forever preferred_lft forever
    inet6 fe80::855:283b:e348:82a0/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
hanz@debian:~$

```

8. Percobaan PING dari PC client router B

```

hanz@debian:~$ ping 192.168.10.1
PING 192.168.10.1 (192.168.10.1) 56(84) bytes of data:
64 bytes from 192.168.10.1: icmp_seq=1 ttl=64 time=1.60 ms
64 bytes from 192.168.10.1: icmp_seq=2 ttl=64 time=1.49 ms
^C
--- 192.168.10.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 1.490/1.545/1.600/0.055 ms
hanz@debian:~$ ping 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data:
64 bytes from 10.10.10.1: icmp_seq=1 ttl=63 time=1.90 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=63 time=1.68 ms
^C
--- 10.10.10.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 1.680/1.788/1.896/0.108 ms
hanz@debian:~$ ping 10.10.10.2
PING 10.10.10.2 (10.10.10.2) 56(84) bytes of data:
64 bytes from 10.10.10.2: icmp_seq=1 ttl=64 time=1.52 ms
^C
--- 10.10.10.2 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 1.519/1.519/1.519/0.000 ms
hanz@debian:~$ ping 10.56.145.67
PING 10.56.145.67 (10.56.145.67) 56(84) bytes of data:
64 bytes from 10.56.145.67: icmp_seq=1 ttl=63 time=1.97 ms
64 bytes from 10.56.145.67: icmp_seq=2 ttl=63 time=2.13 ms
^C
--- 10.56.145.67 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 1.969/2.047/2.125/0.078 ms
hanz@debian:~$

```

9. Konfigurasi blokir ping dari router A ke router B

Firewall								
Filter Rules		NAT	Mangle	Raw	Service Ports	Connections	Address Lists	Layer7 Protocols
+ - ✓ ✗ 📄 🔍		Reset Counters		Reset All Counters				
#	Action	Chain	Src. Address	Dst. Address	Proto...	Src. Port	Dst. Port	In. Int
0	✗ drop	input	10.10.10.1		1 (ic...			

10. Percobaan ping dari router A ke router B dan PC client router B


```

Terminal <1>

[Tab]          Completes the command/word. If the input is ambiguous,
                a second [Tab] gives possible options

/              Move up to base level
..            Move up one level
/command      Use command at the base level
[admin@MikroTik] > ping 10.10.10.2
  SEQ HOST                SIZE TTL TIME  STATUS
    0 10.10.10.2          56  63 1ms  timeout
    1 10.10.10.2          56  63 1ms  timeout
    2 10.10.10.2          56  63 1ms  timeout
    3 10.10.10.2          56  63 1ms  timeout
    4 10.10.10.2          56  63 1ms  timeout
  sent=5 received=0 packet-loss=100%

[admin@MikroTik] > ping 192.168.10.2
  SEQ HOST                SIZE TTL TIME  STATUS
    0 192.168.10.2        56  63 1ms
    1 192.168.10.2        56  63 1ms
    2 192.168.10.2        56  63 2ms
  sent=3 received=3 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms

[admin@MikroTik] >

```

11. Konfigurasi blokir ping ke PC client router B

Firewall									
Filter Rules		NAT	Mangle	Raw	Service Ports	Connections	Address Lists	Layers	
						Reset Counters		Reset All Counters	
#	Action	Chain	Src. Address	Dst. Address	Proto...	Src. Port	Dst. Port		
0 X	drop	input	10.10.10.1		1 (ic...				
1	drop	forward	10.10.10.1		1 (ic...				

12. Percobaan ping dari router A ke PC client router B

```

Terminal <1>

[admin@MikroTik] > ping 192.168.10.2
    SEQ HOST                      SIZE TTL TIME  STATUS
    0 192.168.10.2                  56  63 1ms
    1 192.168.10.2                  56  63 1ms
    2 192.168.10.2                  56  63 2ms
    sent=3 received=3 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms

[admin@MikroTik] > ping 10.10.10.2
    SEQ HOST                      SIZE TTL TIME  STATUS
    0 10.10.10.2                   56  64 0ms
    1 10.10.10.2                   56  64 0ms
    2 10.10.10.2                   56  64 0ms
    sent=3 received=3 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=0ms

[admin@MikroTik] > ping 192.168.10.2
    SEQ HOST                      SIZE TTL TIME  STATUS
    0 192.168.10.2                  56  63 timeout
    1 192.168.10.2                  56  63 timeout
    2 192.168.10.2                  56  63 timeout
    3 192.168.10.2                  56  63 timeout
    sent=4 received=0 packet-loss=100%

[admin@MikroTik] >

```

13. Konfigurasi port forwarding di router B agar PC client router A dapat mengakses aplikasi di PC client di bawah router B (NAT).

Firewall										
Filter Rules NAT Mangle Raw Service Ports Connections Address Lists Layer7 Protocols										
+ - ✓ ✗ 📄 🔍 Reset Counters 00 Reset All Counters										
#	Action	Chain	Src. Address	Dst. Address	Proto...	Src. Port	Dst. Port	In. Inte...	Out. Int...	In
0	mas...	srcnat							ether1	
1	dst...	dstnat		10.10.10.2	6 (tcp)		8080			

14. Nyalakan aplikasi HTTP sederhana di PC client router B.

```

-- sudo -- Konsole
go: bash x station: bash x -- sudo x
-- bash
0194:fxme:lcon:NTUserDrawIconEx Error retrieving icon frame 0
0194:fxme:lcon:NTUserDrawIconEx Error retrieving icon frame 0
0194:fxme:lcon:NTUserDrawIconEx Error retrieving icon frame 0
-- sudo
hanz@debian:~/Documents/Books$ sudo python3 -m http.server 80
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...
10.10.10.1 - - [01/Jun/2026 10:16:20] "GET / HTTP/1.1" 200 -
10.10.10.1 - - [01/Jun/2026 10:17:28] "GET / HTTP/1.1" 200 -

```

15. Lakukan fetch dari PC client router A ke IP router B.

```
Terminal <1>
a second [Tab] gives possible options

/          Move up to base level
..         Move up one level
/command   Use command at the base level
[admin@MikroTik] > /tool fetch url="http://10.10.10.2:8080" keep-result=no
failure: invalid URL
[admin@MikroTik] > /tool fetch url="http://10.10.10.2:8080" keep-result=no
failure: invalid URL
[admin@MikroTik] > /tool fetch url="http://10.10.10.1:8080" keep-result=no
failure: invalid URL
[admin@MikroTik] > /tool fetch url="http://1.1.1.1" keep-result=no
failure: invalid URL
[admin@MikroTik] > /tool fetch url="10.10.10.2:8080" keep-result=no
failure: invalid URL protocol
[admin@MikroTik] > /tool fetch url="http://10.10.10.2:8080" keep-result=no
failure: invalid URL
[admin@MikroTik] > /tool fetch url="http://10.10.10.2:8080/" keep-result=no
status: finished
downloaded: 0KiB-C-z pause]
total: 0KiB
duration: 1s

[admin@MikroTik] >
```

16. Analisis traffic dari router B hasil fetch di atas.

Firewall								
Filter Rules NAT Mangle Raw Service Ports Connections Address Lists Layer7 Protocols								
Tracking								
	Src. Address	/	Dst. Address	Proto...	Connecti...	Timeout	TCP State	Orig./Repl. Rate
SACd	10.10.10.1:43448		10.10.10.2:8080	6 (tcp)		00:00:05 time wait	0 bps/0 bps	407 B/1289 B
C	192.168.10.2:59285		192.168.10.1:20561	17 (u...		00:00:09	1600 bps/0 bps	287.1 KiB/0 B

2 Analisis Hasil Percobaan

Pada praktikum ini, terdapat dua buah router yang masing-masing terhubung satu sama lain. Namun, hanya satu router yang terhubung dengan internet yaitu Router A. Sedangkan, Router B digunakan sebagai NAT dan firewall bagi PC client B. Untuk mengakses konten yang ada di PC client B melalui terminal PC client A maupun Router A dibutuhkan otoritas dari firewall Router B. Dalam praktikum ini diterapkan pemblokiran ICMP kepada Router B dan PC client B, juga dilakukan konfigurasi port forwarding agar service di dalam PC client B dapat diakses menggunakan IP Router B.

3 Hasil Tugas Modul

Ada di README github:

<https://github.com/HanzCEO/Praktikum-Jarkom-2026-Kelompok-25/blob/main/Modul-4/>

4 Kesimpulan

Praktikum ini memvalidasi implementasi kebijakan firewall dan NAT melalui simulasi topologi jaringan. Pengujian mencakup konfigurasi source NAT agar klien internal dapat mengakses jaringan eksternal, serta destination NAT yang berfungsi membelokkan trafik publik secara spesifik menuju server lokal.

Filter firewall diimplementasikan menggunakan chain input untuk mendrop paket yang mengarah ke router, sedangkan chain forward diterapkan untuk membatasi lalu lintas host-to-host antar segmen jaringan. Selama proses tersebut, fungsionalitas connection tracking diamati secara langsung pada router untuk memverifikasi mekanisme pelacakan port dan translasi alamat IP saat komunikasi sesi TCP maupun ICMP berlangsung.

5 Lampiran

5.1 Dokumentasi saat praktikum



5.2 Hasil Challenge Modul

n/a